

WAREHOUSE MANAGEMENT SYSTEM

Operations & Knowledge Handbook

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Purpose	Reference source for WMS operations and text-based RAG

Project Documentation Notice

This handbook describes the fictional Warehouse Management System created as part of the EIGI project. It is intended as a software demonstration and RAG knowledge source. Operational policies described here apply only to this project environment.

1. About the Warehouse Management System

The Warehouse Management System (WMS) is a centralized platform for managing warehouse inventory, products, sellers, receiving, orders, reservations, fulfillment, barcode scanning, users, and operational audit records.

The system supports two warehouse locations: **Reno** and **Columbus**.

The platform provides a unified interface for warehouse managers, inventory staff, administrators, and read-only users.

2. Core Warehouse Concepts

Product: A product represents an item maintained in the product catalog. A product can contain a product name, SKU, UPC, seller, and active/inactive status.

SKU: The internal stock-keeping identifier used to uniquely identify a product within the warehouse system.

UPC: A product barcode identifier. UPC values are treated as strings so that leading zeros are preserved.

What is a SKU?

A SKU is the internal stock-keeping identifier used to uniquely identify a product within the warehouse system.

What is a UPC?

A UPC is a product barcode identifier. The system treats UPC values as strings so that leading zeros are preserved.

```
012345678905
must remain
012345678905
```

3. Warehouses

The current WMS environment contains two warehouse locations: **Reno** and **Columbus**.

Each warehouse maintains its own inventory position independently. Inventory queries should identify the warehouse whenever a user is asking about a specific location.

How do I check stock in Reno?

Search for the product or SKU and view its Reno inventory position.

How do I check stock in Columbus?

Search for the product or SKU and view its Columbus inventory position.

Does inventory in Reno automatically equal inventory in Columbus?

No. Each warehouse maintains its own inventory position.

4. Inventory Management

Inventory represents the quantity of a product available within a warehouse.

The system supports inventory lookup, stock adjustments, reservations, inventory movement history, and warehouse-specific inventory queries.

Inventory changes should be performed through the WMS workflow rather than by directly modifying database records.

4.1 Inventory Adjustment

SKU: SKU-1048
Warehouse: Reno
Adjustment: +10

How do I adjust inventory?

Open the inventory record, select the appropriate warehouse and product, enter the quantity adjustment, and submit the operation.

Can a read-only user adjust inventory?

No. Read-only users cannot perform inventory mutations.

What happens after an inventory adjustment?

The inventory position is updated and the operational action is recorded in the audit trail.

Why might an inventory adjustment fail?

An adjustment may fail because of invalid input, insufficient permissions, conflicting inventory state, or another backend validation error.

5. Inventory Reservations

A reservation temporarily allocates inventory for an order. Reservations help prevent the same inventory from being committed to multiple orders.

The reservation workflow is protected against concurrent operations.

What does reserving inventory mean?

It means allocating available stock to an order so that the quantity can be used for that order's fulfillment.

Can I reserve more inventory than is available?

No. The system validates inventory availability before completing a reservation.

What happens if two users try to reserve the same stock simultaneously?

The backend uses concurrency protection to prevent inconsistent inventory reservations.

6. Inventory Movements

Inventory movements provide a history of stock changes. A movement may be associated with receiving, adjustment, reservation, or other inventory operations.

The movement history helps warehouse personnel understand why the current inventory quantity exists.

How can I investigate a stock change?

Review the inventory movement history and the corresponding audit events.

7. Product Management

The product module maintains the product master. Authorized users can view products, search products, create products, and look up products by SKU or UPC.

Inactive products remain identifiable but should not be treated as active inventory items for normal operations.

Can I find a product using its SKU?

Yes.

Can I find a product using its UPC?

Yes.

Does the system preserve leading zeros in UPCs?

Yes. UPCs are treated as strings.

8. Barcode Scanning

The WMS supports barcode-based product identification. A barcode can be entered manually or scanned using a keyboard-emulating USB barcode scanner.

Scanning normalizes the input, resolves the product, and can display product information together with Reno and Columbus stock.

```
Barcode
↓
Normalization
↓
Product lookup
↓
Product identification
↓
Warehouse inventory lookup
```

Can I use a physical barcode scanner?

Yes. The scanner interface supports keyboard-emulating USB barcode scanners.

What happens if a barcode is not found?

The system returns a product-not-found response rather than guessing a product.

Are leading zeros preserved?

Yes.

9. Seller Management

Sellers represent organizations or parties associated with products. Authorized users can view sellers, search sellers, create sellers, and review seller-related product information.

Can a product be associated with a seller?

Yes. Products contain seller information where applicable.

Who can create sellers?

The operation is controlled by the user's RBAC permissions.

10. Receiving

Receiving records the arrival of inventory into a warehouse. A receiving operation may contain information about an incoming shipment, products, quantities, warehouse, and discrepancies.

Receiving is an important point at which expected inventory is compared with physically received inventory. Discrepancies should be recorded rather than silently hidden.

What is receiving?

Receiving is the process of recording inventory that arrives at a warehouse.

What should happen when received quantities do not match the expected quantities?

The discrepancy should be recorded rather than silently modifying inventory to hide the difference.

Where can I review receiving activity?

Use the Receiving module and the Audit Trail.

11. Orders

The order module manages customer or downstream warehouse orders. The system supports viewing, searching, and creating orders and tracking their status.

How do I create an order?

Open the Orders module, provide the required order information and line items, and submit the order.

What happens after an order is created?

The order enters the appropriate operational state and can proceed through reservation and fulfillment.

12. Fulfillment

Fulfillment manages the physical processing of an order.

Picking identifies and collects the inventory required for an order. Packing prepares the picked order for shipment. Shipping marks the order as dispatched.

```
Confirmed
↓
Reserved
↓
Picking
↓
Packed
↓
Shipped
```

What is the fulfillment sequence?

The standard sequence is Confirmed → Reserved → Picking → Packed → Shipped.

Can I ship an order before it is packed?

The fulfillment workflow enforces the appropriate state progression.

Why can a fulfillment action fail?

It may fail because the order is in an invalid state, inventory requirements are not satisfied, the user lacks permission, or another validation rule prevents the operation.

13. Audit Trail

The Audit Trail records important operational actions performed within the WMS. Audit records can contain action, entity type, entity ID, user, warehouse, previous state, new state, metadata, success/failure, and timestamp.

The audit trail provides traceability for warehouse operations.

Why does the WMS have an audit trail?

To provide a historical record of important system operations and state changes.

Can users edit an audit record?

Audit records are intended to be immutable operational history.

Can I search audit activity?

Yes. Audit records can be filtered by information such as action, entity, user, warehouse, success state, and time range.

Are inventory adjustments audited?

Yes. Operational mutations such as inventory adjustments generate audit activity.

14. Users and Roles

Access is controlled using authentication and role-based access control.

Supported roles include **ADMIN**, **WAREHOUSE_MANAGER**, **INVENTORY_CLERK**, and **READ_ONLY**.

The backend remains the authoritative enforcement layer for permissions.

Can every user perform every operation?

No. Operations depend on the permissions associated with the user's role.

Can a **READ_ONLY** user adjust inventory?

No.

Can frontend permissions replace backend authorization?

No. Frontend restrictions improve usability, but the backend must enforce authorization.

15. Authentication

Users must authenticate before accessing protected WMS functionality.

The authentication flow obtains a JWT access token and uses the authenticated session to access protected WMS APIs.

```
Username / Credentials
↓
Authentication API
↓
JWT Access Token
↓
Authenticated Session
↓
Protected WMS APIs
```

What happens after login?

The system obtains an access token and retrieves the authenticated user's information.

What happens if the session expires?

The frontend should clear the invalid session and require the user to authenticate again.

16. Voice AI

The WMS includes a browser-based voice interface for operational commands. Voice input is converted into text using browser speech recognition.

Voice commands are converted into structured intents and entities, validated against RBAC permissions, and dispatched through normal WMS APIs.

Mutating voice commands require confirmation before execution. Ambiguous commands should request clarification rather than guessing.

```
graph TD; A[Microphone] --> B[Speech-to-Text]; B --> C[Transcript]; C --> D[Intent recognition]; D --> E[Entity extraction];
```

Microphone
↓
Speech-to-Text
↓
Transcript
↓
Intent recognition
↓
Entity extraction

↓
RBAC validation
↓
WMS API

Can voice commands directly access MongoDB?

No. Voice commands use the normal backend API and business-logic layers.

Can voice commands perform inventory mutations?

Authorized users can initiate supported mutations, but the system requires confirmation before executing them.

Can an ambiguous voice command be executed automatically?

No. Missing or ambiguous required information should result in clarification rather than guessing.

17. Knowledge Center and RAG

The Knowledge Center is the text-based knowledge interface for WMS documentation.

Knowledge questions should use the handbook and other indexed documentation as their source of truth.

Operational questions requiring live values, such as current stock or order status, should use live WMS APIs rather than fabricated RAG answers.

User Question
↓
Query Processing
↓
Embedding
↓
Vector Search
↓
Relevant Handbook Chunks
↓
LLM
↓
Answer + Sources

What is RAG used for?

RAG is used to answer text-based questions from the WMS knowledge base and operational handbook.

Does RAG replace live inventory queries?

No. Live operational data should come from the WMS APIs.

Can RAG answer questions that are not in the handbook?

The system should clearly indicate when the available knowledge sources do not contain sufficient information rather than inventing an answer.

18. Frequently Asked Questions

What is this system?

It is a warehouse management platform for managing products, sellers, inventory, receiving, orders, fulfillment, users, barcode scanning, and operational audit records.

Which warehouses are supported?

The current system supports Reno and Columbus.

Can I access the system without logging in?

Protected WMS functionality requires authentication.

How do I check stock?

Use the Inventory module and select or search for the relevant product.

Can I check stock by SKU?

Yes.

Can I check stock by barcode?

Yes. Use the Scanner module.

Can I adjust stock?

Authorized users can perform inventory adjustments.

Are stock adjustments recorded?

Yes, operational changes are recorded in the audit trail.

Does the scanner support UPCs with leading zeros?

Yes.

Can I use a USB scanner?

Yes, provided it behaves as a keyboard-emulating barcode scanner.

What happens when a barcode doesn't exist?

The system reports that the product could not be found.

Where do I record incoming stock?

Use the Receiving module.

Where can I investigate a receiving discrepancy?

Review the receiving record and corresponding audit history.

How does an order move through fulfillment?

Orders progress through reservation, picking, packing, and shipping according to the workflow.

Can I skip fulfillment stages?

The backend enforces valid workflow transitions.

Can a read-only user modify inventory?

No.

Where is authorization enforced?

Authorization is enforced by the backend through authentication and RBAC.

How can I see who changed inventory?

Use the Audit Trail to search the relevant inventory activity.

What is RAG used for?

RAG is used for text-based questions grounded in the WMS handbook and indexed documentation.

19. Operational Policies

Code	Policy
WMS-PROD-01	Product and SKU Management
WMS-INV-02	Inventory Management
WMS-RES-03	Inventory Reservation

Code	Policy
WMS-RCV-04	Receiving Operations
WMS-ORD-05	Order Management
WMS-FUL-06	Fulfillment Operations
WMS-BC-07	Barcode and UPC Scanning
WMS-AUD-08	Audit Trail
WMS-RBAC-09	Authentication and Role-Based Access Control
WMS-RAG-10	Knowledge Center and RAG
WMS-VAI-11	Voice Command Operations

20. RAG Knowledge Boundaries

Rule 1: Use the handbook as the source of truth

If an answer is not supported by the knowledge base, the system should say so.

Rule 2: Do not fabricate warehouse data

RAG should not invent stock quantities, order status, product availability, user information, or audit events.

Rule 3: Separate knowledge from operations

"How does inventory reservation work?" → RAG

"Reserve 5 units for ORD-1001." → WMS API

Rule 4: Show sources

RAG answers should ideally expose the handbook section or document chunk used to generate the response.

Rule 5: Keep the RAG text-based

Text → RAG → Answer

Voice-to-RAG is not part of this phase.

21. Document Version & Maintenance

Document: Warehouse Management System Operations & Knowledge Handbook

Version: 1.0

Status: Initial RAG knowledge source

Review: Update whenever a warehouse workflow, permission model, API behavior, or operational policy changes.